

SOFTWARE ENGINEERING PROJECT

HOSTELFIT

Mess calorie & workout tracker

A hostel-focused nutrition and fitness tracking web application

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PROJECT

Software Engineering Project

TEAM

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CONTEXT

HOSTELFIT CONNECTS THE DAILY HOSTEL ROUTINE TO FITNESS PROGRESS

Hostel students often work within fixed mess menus, while generic calorie apps make them search manually and separate meal, workout, and weight tracking. HostelFit brings those actions into one workflow.

01**Hostel Mess**

Start with the fixed hostel menu.

02**Meal Logging**

Record selected meals quickly.

03**Calorie Analysis**

Estimate intake and expenditure.

04**Fitness Progress**

Follow weight and progress over time.

■ **TAKEAWAY:** HostelFit joins four daily actions that generic tools keep apart.

PROBLEM

THE PROBLEM IS FRAGMENTED TRACKING AROUND A FIXED MENU

WHY TRACKING BREAKS

- Fixed hostel and mess menus make calorie estimation difficult.
- Food intake and physical activity are rarely integrated in one hostel-specific system.
- Generic fitness applications do not reflect the hostel context.

WHAT HOSTELFIT SHOULD ENABLE

- Pre-load Mess Menu data and support quick Meal Logging.
- Estimate Calorie Intake and Workout Calorie Expenditure.
- Support personalized targets plus Weight and Progress tracking.

■ **TAKEAWAY:** The system is designed around the hostel menu, not around repeated food search.

METHOD

AN INCREMENTAL DEVELOPMENT PATH TURNED REQUIREMENTS INTO A PROTOTYPE

Requirements were translated into UML Use Case, DFD Level 0, DFD Level 1, DFD Level 2, Activity, and ER diagrams. Modules were implemented and integrated progressively.

1 Requirements Analysis	2 System Design	3 Database Design	4 Frontend Development
5 Backend Development	6 System Integration	7 Testing & Validation	8 Prototype Evaluation

WORKFLOW

THE CORE WORKFLOW MOVES FROM PROFILE SETUP TO MEASURABLE DAILY BALANCE

01 User Registration → Profile Setup → Fitness Goal Selection

02 Hostel/Mess Selection → View Mess Menu → Meal Logging

03 Calorie Intake Calculation → Workout Logging → Calorie Expenditure Estimation

04 Daily Calorie Balance → Progress Tracking

SEPARATE ADMIN PATH

ADMIN

Manage Mess Menu

■ **TAKEAWAY:** The user path measures daily balance; the Admin path maintains the menu foundation.

ARCHITECTURE

THREE LAYERS SEPARATE THE INTERFACE, APPLICATION LOGIC, AND DATA

PRESENTATION LAYER

Web Interface · Login / Registration · Dashboard · Meal Logging · Workout Logging · Progress Tracking

APPLICATION LAYER

Authentication · Profile & Goal Management · Menu Management · Meal Processing · Workout Processing · Calorie Calculations · Progress Analysis

DATA LAYER

Users · **Profiles** · **Fitness Goals** · **Hostel/Mess** · **Menu Items** · Meal Records · Workout Records · Weight & Progress Data

Frontend

Backend API

Database

PROTOTYPE

THE PROTOTYPE MAKES THE PLANNED HOSTELFIT WORKFLOW TANGIBLE

Clean browser-style panels show the main modules while keeping the scope grounded in prototype functionality.

HOSTELFIT / DASHBOARD

User Profile & Fitness Goal Selection

Registration / Login

Goal selection · weight record · daily overview

MESS MENU

Meal Logging

Hostel/Mess Menu

Selected meal · estimated Calorie Intake

FITNESS LOG

Workout Logging & Calorie Balance

Workout record · estimated Calorie Expenditure

Weight & Progress Tracking

ADMIN

Menu Management

Prototype functionality for maintaining mess menu data.

VALIDATION

THE PROTOTYPE VALIDATES THE INTEGRATED WORKFLOW, NOT PERFORMANCE CLAIMS

WHAT THE PROTOTYPE SHOWS

- Core HostelFit workflow was translated into functional prototype modules.
- Pre-loaded mess data reduces repeated manual food searching.
- Meal records support estimated daily Calorie Intake; Workout records support estimated Calorie Expenditure.
- Weight and activity information can be organized for Progress monitoring.

FEATURE STATUS

Feature	Status
User Authentication	Implemented
Profile & Goals	Implemented
Mess Menu	Implemented
Meal Logging	Implemented
Workout Logging	Implemented
Calorie Balance	Implemented
Progress Tracking	Implemented
Admin Menu Management	Prototype functionality

CONCLUSION

A FEASIBLE FOUNDATION

HostelFit demonstrates a feasible foundation for hostel-centered fitness tracking.

Mess-menu selection · meal and workout logging · calorie balance · progress tracking

CONCLUSION & FUTURE SCOPE

WHAT THE PROJECT ESTABLISHES

- Addresses the nutrition and fitness tracking needs of hostel students.
 - Connects mess-menu selection, meal and workout logging, calorie balance, and progress tracking.
 - Builds on UML, DFD, Activity, and ER modelling.
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WHAT COMES NEXT

- PostgreSQL-backed production database
- Secure authentication and role-based authorization
- Better food-portion estimation and more hostel/mess datasets
- Wearable-device integration, notifications, reminders, and advanced analytics